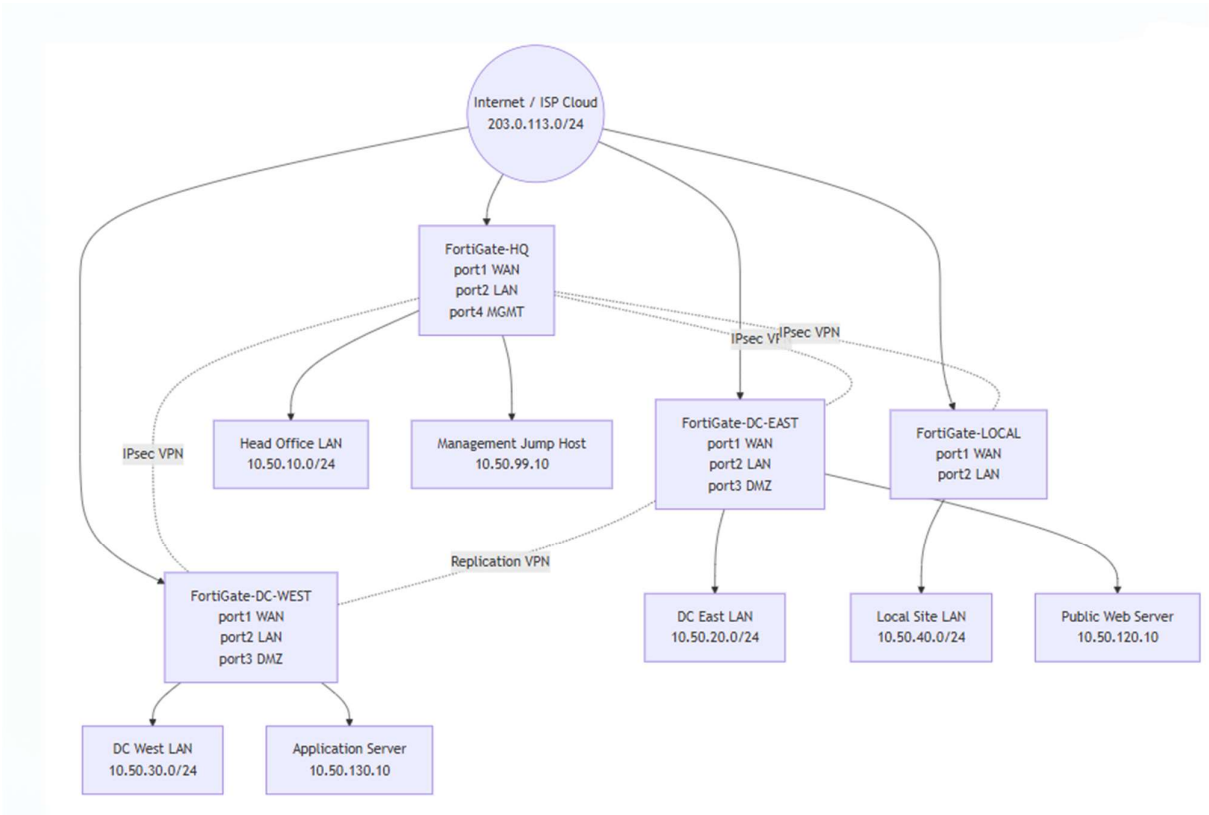


Electric Petrol Perimeter Design

Purpose

This submission provides a perimeter design for Electric Petrol using FortiGate firewalls. It includes Internet connectivity, site-to-site VPNs, DMZ services, and remote access controls.

Topology Diagram



Addressing Plan

Zone	Network	Gateway / Host
ISP / Internet	203.0.113.0/24	203.0.113.1
Head Office LAN	10.50.10.0/24	10.50.10.1
HQ Management	10.50.99.0/24	10.50.99.1
DC East LAN	10.50.20.0/24	10.50.20.1
DC East DMZ	10.50.120.0/24	10.50.120.1
DC West LAN	10.50.30.0/24	10.50.30.1
DC West DMZ	10.50.130.0/24	10.50.130.1
Local Site LAN	10.50.40.0/24	10.50.40.1
SSL VPN Pool	10.50.250.0/24	Assigned by FortiGate-HQ

Internet Connectivity

Each FortiGate uses port1 as the WAN interface and has a default route to 203.0.113.1. LAN-to-Internet firewall policies use NAT.

Site-to-Site VPN Design

Tunnel	Purpose
HQ-to-DCE	Headquarters to DC East LAN and DMZ
HQ-to-DCW	Headquarters to DC West LAN and DMZ
HQ-to-LOCAL	Headquarters to Local Site LAN
DCE-to-DCW	Data centre replication / DR traffic

The VPNs use IKEv2, AES256/SHA256, DH group 14, and NAT is disabled on VPN policies.

DMZ Services

Service	Location	DMZ IP	Public IP / VIP
Public Web Server	DC East DMZ	10.50.120.10	203.0.113.20 HTTP/HTTPS
Application Server	DC West DMZ	10.50.130.10	203.0.113.30 HTTPS

DMZ services are separated from the internal LANs. Public users can access only published VIP services.

Results Section

Paste actual command output/screenshots here after testing.

Internet Test

execute ping 203.0.113.1

Expected: successful replies.

VPN Tunnel Test

get vpn ipsec tunnel summary

diagnose vpn tunnel list

Expected: tunnels up and counters increasing.

Site-to-Site Ping Test

execute ping-options source 10.50.10.1

execute ping 10.50.20.1

execute ping 10.50.30.1

execute ping 10.50.40.1

Expected: all successful.

DMZ Tests

curl http://203.0.113.20

curl https://203.0.113.20

curl https://203.0.113.30

Expected: published DMZ services respond.

Negative Security Tests

ping 10.50.20.100

ping 10.50.30.100

curl http://10.50.120.10

Expected: blocked from Internet side.